

DEPARTMENT OF STATISTICAL SCIENCE
2026 - ADVANCED REGRESSION TOPICS
TUTORIAL EXERCISES

1. Consider the linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where $\mathbf{y}$ is an N -dimensional vector, $\mathbf{X}$ is an $N \times (p+1)$ dimensional design matrix, and $\boldsymbol{\varepsilon}$ is an N -dimensional random vector with mean $\mathbf{0}$ and covariance matrix $\sigma^2 \mathbf{I}$.

- (a) Show that minimising the residual sum of squares is equivalent to finding the residual vector $\hat{\boldsymbol{\varepsilon}}$ such that $\hat{\boldsymbol{\varepsilon}} \perp \mathcal{C}(\mathbf{X})$ ($\mathcal{C}(\mathbf{X})$ is the column space of $\mathbf{X}$).
- (b) Show that $\frac{(N-p-1)\hat{\sigma}^2}{\sigma^2} \sim \chi_{N-p-1}^2$ where $\hat{\sigma}^2 = \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N-p-1}$.
- (c) Show by derivation how the mean squared error of the least squares estimator $\hat{\boldsymbol{\beta}}$ can be decomposed into its two components, and comment on the relevance of this decomposition for model selection.
- (d) Consider fitting the linear model with N observations and $p+1$ regression coefficients subject to the constraints $\mathbf{A}\boldsymbol{\beta} = \mathbf{C}$, where $\mathbf{A}$ is a $q \times (p+1)$ matrix of rank $(p+1)$ and $\mathbf{C}$ is a vector of constants. Show that the constrained least squares estimator of the coefficients is given by

$$\hat{\boldsymbol{\beta}}_C = \hat{\boldsymbol{\beta}} - (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{A}^T [\mathbf{A}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{A}^T]^{-1} (\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{C})$$

where $\hat{\boldsymbol{\beta}}$ is the least squares estimator for the unconstrained model.

- (e) Show that the difference between the residual sum of squares for the constrained and unconstrained models is

$$RSS_C - RSS = (\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{C})^T [\mathbf{A}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{A}^T]^{-1} (\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{C})$$

- (f) Hence, show that

$$F = \frac{(RSS_C - RSS)/q}{RSS/(N-p-1)}$$

has an $F_{q, N-p-1}$ distribution under the null hypothesis $\mathbf{A}\boldsymbol{\beta} = \mathbf{C}$. State any assumptions that are required in the derivation.

2. Consider the following two linear regression models

$$\text{Model 1} \quad \mathbf{Y}_{(n \times 1)} = \beta_0 \mathbf{1} + \boldsymbol{\varepsilon}_{(n \times 1)}$$

$$\text{Model 2} \quad \mathbf{Y}_{(n \times 1)} = \mathbf{X}_{(n \times 1)} \boldsymbol{\beta} + \boldsymbol{\varepsilon}_{(n \times 1)}$$

where $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ and $\mathbf{1}$ is an n -dimensional vector of ones. Define $p = k - 1$, $\hat{\boldsymbol{\beta}}$ as the ordinary least squares estimate of the $\boldsymbol{\beta}$ coefficients of the full regression model

(Model 2), $\bar{y}$ as the sample mean of the response variable and $\hat{\beta}^{(1)}$ as a p -dimensional sub-vector of $\hat{\beta}$ with $\hat{\beta}^{(1)} = (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p)^T$. Consider model 1:

- (a) Show that $\hat{\beta}_0 = \bar{y}$.
- (b) Provide an expression for the estimated residual variance of Model 1.

consider Model 2:

- (c) Show that the residual sum of squares for Model 2 is

$$SSE_2 = \mathbf{Y}^T \mathbf{Y} - \hat{\beta}^T \mathbf{X}^T \mathbf{Y}$$

- (d) Provide the null and the alternate hypotheses in order to assess whether a significant regression model exists.
- (e) Use your above answers to show that the required test statistic when undertaking the above hypothesis test is

$$F = \frac{\hat{\beta}^T \mathbf{X}^T \mathbf{Y} - n\bar{y}}{pS_2^2}$$

- (f) where S_2^2 is the estimated residual variance of Model 2. Explain your reasoning. State any assumptions made where needed.
- (g) What is the distribution of the above test statistic? (no proof required)

3. Let

$$X = \begin{pmatrix} 2 & -5 & 3 & 1 \\ 0 & 4 & -3 & -1 \\ -2 & -2 & 3 & 0 \end{pmatrix}$$

Find an orthonormal basis for $\mathcal{C}(X)$, the column space of X .